

Peano axioms from set theory

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1 Axioms of ZF

The following takes place in first order logic. Note that in first order logic, for any symbols x, y and any formula ϕ with a free variable z , then $x = y \implies (\phi(x) \iff \phi(y))$. Additionally, we must have $x = x$. This is not the definition of equality in first order logic; the implication $\phi(x) \implies \phi(y)$ is one of the axioms according to Wikipedia. However, it can be proven that $x = y$ implies $y = x$, and thus the other direction follows.

Axiom 1 (Extensionality). *Sets are equal if and only if they have the same elements:*

$$\forall x \forall y (\forall z (z \in x \iff z \in y) \implies x = y).$$

Axiom 2 (Regularity). *Sets cannot contain themselves:*

$$\forall x ((\exists a (a \in x)) \implies \exists y (y \in x \wedge \neg \exists z (z \in y \wedge z \in x))).$$

Axiom 3 (Schema of restricted comprehension). *Given any formula φ in the language of ZFC which is free in variables $x, w_1, \dots, w_n, z$ (not y), we have the following axiom, which provides set builder notation:*

$$\forall z \forall w_1 \forall w_2 \dots \forall w_n \exists y \forall x (x \in y \iff ((x \in z) \wedge \varphi(x, w_1, w_2, \dots, w_n, z))).$$

We typically write $y = \{x \in z \mid \varphi(x)\}$.

Axiom 4 (Pairing). *If x, y are sets, then there exists a set containing x and y :*

$$\forall x \forall y \exists z ((x \in z) \wedge (y \in z)).$$

Note that this only supplies a set containing x, y , but does not supply the set $\{x, y\}$.

Axiom 5 (Union). *Unions of families exist:*

$$\forall \mathcal{F} \exists A \forall Y \forall x ((x \in Y \wedge Y \in \mathcal{F}) \implies x \in A).$$

This axiom does not assert the existence of $\cup \mathcal{F}$, but provides a set containing $\cup \mathcal{F}$.

Axiom 6 (Schema of replacement). *Let φ be any formula in the language of ZFC which is free in variables $x, y, A, w_1, \dots, w_n$, and not B . Then we have the axiom*

$$\forall A \forall w_1 \dots \forall w_n (\forall x (x \in A \implies \exists! y \varphi) \implies \exists B \forall x (x \in A \implies \exists y (y \in B \wedge \varphi))).$$

This allows the image of a function to fall inside a set.

Definition 1. Let $\mathcal{F}$ be a set. Let A be the set guaranteed by the axiom of union. Define the set

$$\cup \mathcal{F} := \{x \in A \mid \exists Y ((Y \in \mathcal{F}) \wedge (x \in Y))\}.$$

This exists by the axiom schema of restricted comprehension.

Construction 2. Let x, y be sets. Then by the axiom of pairing, there exists a set z such that $x, y \in z$. Define

$$\{x, y\} := \{a \in z \mid (a = x) \vee (a = y)\}.$$

This formula is well-defined, and thus $\{x, y\}$ exists by the axiom schema of restricted comprehension. We can similarly define a set

$$\{x\} := \{a \in z \mid (a = x)\}.$$

Definition 3. With notation as in the above construction, define

$$x \cup y := \cup \{x, y\}.$$

Axiom 7 (Infinity). *A set exists, and it contains the naturals up to isomorphism. Here we write $\emptyset$ for the empty set. The axiom can be modified to postulate the existence of $\emptyset$ as well as the set X , but we present a simpler version using two axioms. Write $s(x)$ for the set $x \cup \{x\}$. This set exists by the above definition, and by modifying the construction above it to select only the element x .*

$$\exists \emptyset \forall x (\neg(x \in \emptyset)).$$

$$\exists X ((\emptyset \in X) \wedge (x \in X \implies s(x) \in X)).$$

Definition 4 (Subset). Let x, y be sets. Then we define

$$(y \subseteq x) \iff \forall z (z \in y \implies z \in x).$$

Axiom 8 (Power set). *Power sets exist:*

$$\forall x \exists y \forall z (z \subseteq x \implies z \in y).$$

The set y contains the power set of x . We then define

$$\mathcal{P}(x) := \{z \in y \mid z \subseteq x\}.$$

We call $\mathcal{P}(x)$ the power set of x .

Remark 5. The axiom of choice requires us to define functions. Since functions are subsets of cartesian products, we would need to define cartesian products. This is not necessary for our purposes. Thus, we work in ZF, rather than ZFC. This has the advantage of being simpler, since C is independent of ZF.

2 Definition of $\mathbb{N}$ and the Peano axioms

Definition 6. Let x be a set. We define the *successor* of x , denoted $s(x)$, to be the set $x \cup \{x\}$.

Definition 7. Let X be a set. We say X is an *inductive set* if $\emptyset \in X$ and, for all $x \in X$, we have $s(x) \in X$.

Note that by the [axiom of infinity](#), there exists at least one inductive set. Let $\mathcal{X}$ denote this set.

Construction 8. By the [axiom of power set](#), the power set $\mathcal{P}(\mathcal{X})$ exists. Let

$$\mathcal{I} := \{X \in \mathcal{P}(\mathcal{X}) \mid X \text{ is inductive}\}.$$

Since being inductive is a well-defined formula (X is inductive if and only if $(\emptyset \in X) \wedge \forall x (x \in X \implies s(x) \in X)$), this set $\mathcal{I}$ exists by the [axiom schema of restricted comprehension](#).

Definition 9. Let $\mathcal{F}$ be a set. Define the *intersection* $\cap \mathcal{F}$ to be the set

$$\cap \mathcal{F} := \{x \in \cup \mathcal{F} \mid \forall X (X \in \mathcal{F} \implies x \in X)\}.$$

Definition 10. Define the set $\mathbb{N} := \cap \mathcal{I}$. We call $\mathbb{N}$ the *natural numbers*.

Lemma 11. *Let X be a set. Then $X \in \mathcal{P}(X)$.*

Proof. Note that $x \in X$ implies $x \in X$, and thus $X \subseteq X$ by definition 4. Let y be the set guaranteed by the [axiom of power set](#), such that $\mathcal{P}(X) \subseteq y$. Since $X \subseteq X$, we have $X \in y$ by the [axiom of power set](#). Since $X \subseteq X$, we have $X \in \mathcal{P}(X)$ by the [axiom schema of restricted comprehension](#). ■

Lemma 12. *Let $\mathcal{F}$. For all $X \in \mathcal{F}$ and all $x \in X$, we have $x \in \cup \mathcal{F}$.*

Proof. Let $\mathcal{F}$ be a set, and let $X \in \mathcal{F}$ and $x \in X$. By the [axiom of union](#), there is a set A such that $(x \in X) \wedge (X \in \mathcal{F})$ implies $x \in A$. By definition 1, $\cup \mathcal{F}$ is the subset of A guaranteed by the [axiom schema of restricted comprehension](#) such that

$$\cup \mathcal{F} = \{x \in A \mid \exists X((X \in \mathcal{F}) \wedge (x \in X))\}.$$

Hence, $x \in X$ and $X \in \mathcal{F}$ implies $x \in \cup \mathcal{F}$ by the [axiom schema of restricted comprehension](#). Thus, $x \in \cup \mathcal{F}$. ■

Lemma 13. *Let $\mathcal{F}$ be a nonempty set. Then $x \in \cap \mathcal{F}$ if and only if for all $X \in \mathcal{F}$, we have $x \in X$. Symbolically,*

$$(x \in \cap \mathcal{F}) \iff \forall X(X \in \mathcal{F} \implies x \in X).$$

Proof. ($\implies$) Let $x \in \cap \mathcal{F}$. Then by definition 9 and the [axiom schema of restricted comprehension](#), for all $X \in \mathcal{F}$, we have $x \in X$.

($\impliedby$) Let $x \in X$ for all $X \in \mathcal{F}$. Then there is some $X \in \mathcal{F}$ such that $x \in X$, and by lemma 12 we have $x \in \cup \mathcal{F}$. By the [axiom schema of restricted comprehension](#), since $\forall X(X \in \mathcal{F} \implies x \in X)$ and $x \in \cup \mathcal{F}$, we have $x \in \cap \mathcal{F}$. ■

Lemma 14. *The set $\mathbb{N}$ is inductive.*

Proof. Let $\varphi(x)$ if and only if $\forall X(X \in \mathcal{J} \implies x \in X)$. By the [axiom schema of restricted comprehension](#) and definition 10, we have $x \in \mathbb{N}$ if and only if $x \in \cup \mathcal{J}$ and $\varphi(x)$.

Note that $\mathcal{X}$ is an inductive set by the [axiom of infinity](#). Thus, $\mathcal{X} \in \mathcal{J}$, meaning $\mathcal{X}$ is nonempty. Let $X \in \mathcal{J}$. Then since X is inductive, by definition 7 we have $\emptyset \in X$. Thus, $\forall X(X \in \mathcal{J} \implies \emptyset \in X)$. By lemma 13, $\emptyset \in \cap \mathcal{J}$. Now let $x \in \mathbb{N}$. We must show $s(x) \in \mathbb{N}$ to complete the proof. Since $x \in \mathbb{N} = \cap \mathcal{J}$, by lemma 13, for all $X \in \mathcal{J}$, we have $x \in X$. Let $X \in \mathcal{J}$. Then $x \in X$, and since X is inductive, by definition 7 we have $s(x) \in X$. Thus, for all $X \in \mathcal{J}$, we have $s(x) \in X$. By lemma 13, we have $s(x) \in \cap \mathcal{J} = \mathbb{N}$. Therefore, $\mathbb{N}$ is inductive. ■

Lemma 15. *Let x be a set. Then $x \in s(x)$.*

Proof. Recall construction 2. Let x be a set. Since $\mathcal{X}$ also exists, by the [axiom of pairing](#), there is a set z with $x, \mathcal{X} \in z$. By the [axiom schema of restricted comprehension](#), we can define a set

$$\{x\} := \{a \in z \mid a = x\}.$$

By [axiom schema of restricted comprehension](#), we have $a \in \{x\}$ if and only if $a \in z$ and $a = x$. Note that $a = x$ implies $a \in z$ since $x \in z$, and by definition of equality in first order logic. Thus, we have $a \in \{x\}$ if and only if $a = x$. Now recall that $s(x) = x \cup \{x\}$. Note that by construction 2, we have $\{x\} \in \{x, \{x\}\}$. By lemma 12, since $x \in \{x\} \in \{x, \{x\}\}$, we have

$$x \in \cup \{x, \{x\}\} = x \cup \{x\} = s(x),$$

where the first quality is by definition 3, and the second is by definition 6. Thus, $x \in s(x)$. ■

Lemma 16. *Let $A \subseteq B \subseteq C$. Then $A \subseteq C$.*

Proof. Let $a \in A$. Then $A \subseteq B$ implies $a \in B$. Then $B \subseteq C$ implies $a \in C$. Thus $a \in A$ implies $a \in C$, and thus $A \subseteq C$. ■

Theorem 17 (Peano Axioms). *The following are true.*

1. $\emptyset \in \mathbb{N}$.
2. If $n \in \mathbb{N}$, then $s(n) \in \mathbb{N}$.
3. For all $n \in \mathbb{N}$, $s(n) \neq \emptyset$.
4. If $s(n) = s(m)$, then $n = m$.
5. If $A \subseteq \mathbb{N}$ is such that $\emptyset \in A$ and $n \in A \implies s(n) \in A$, then $A = \mathbb{N}$. In other words, $\mathbb{N}$ has no proper inductive subset.

Proof. (1) By lemma 14, $\mathbb{N}$ is inductive. By definition 7, $\emptyset \in \mathbb{N}$.

(2) By lemma 14, $\mathbb{N}$ is inductive. By definition 7, for all $n \in \mathbb{N}$, we have $s(n) \in \mathbb{N}$.

(3) Let $n \in \mathbb{N}$. Then since $n \in s(n)$ by lemma 15, we have that $s(n) \neq \emptyset$ since $n \in \emptyset$ is false for all n .

(5) Let $A \subseteq \mathbb{N}$ be inductive. Note that by lemma 13, we have $n \in \mathbb{N}$ if and only if $n \in \mathcal{X}$ (since $\mathcal{X} \in \mathcal{J}$). Hence, $\mathbb{N} \subseteq \mathcal{X}$. Since $A \subseteq \mathbb{N} \subseteq \mathcal{X}$, by lemma 16, $A \subseteq \mathcal{X}$. Thus, $A \in \mathcal{P}(\mathcal{X})$ by definition and the [axiom schema of restricted comprehension](#). Since A is inductive and $A \in \mathcal{P}(\mathcal{X})$, by [axiom schema of restricted comprehension](#) and definition of $\mathcal{J}$, we have $A \in \mathcal{J}$. By lemma 13, if $n \in \cap \mathcal{J}$, then $n \in A$. Since $\cap \mathcal{J} = \mathbb{N}$, we have for all $n \in \mathbb{N}$, it follows that $n \in A$. Thus $n \in \mathbb{N} \implies n \in A$. By definition 4, since $A \subseteq \mathbb{N}$, we have $n \in A \implies n \in \mathbb{N}$. Therefore, $n \in A \iff n \in \mathbb{N}$. By the [axiom of extensionality](#), we have $A = \mathbb{N}$.

The proof of Peano axiom (4) requires a lemma which requires Peano axiom (5), and a lemma requiring the axiom of regularity.

Lemma 18. *Let $n, m \in \mathbb{N}$ and $n \in m$. Then $n \subseteq m$.*

Proof. Since we have proven Peano axiom (5), we use it for this proof. Define the set

$$A = \{m \in \mathbb{N} \mid \forall n(n \in m \implies n \subseteq m)\}.$$

We first show that $A = \mathbb{N}$. First, note that by definition we have $A \subseteq \mathbb{N}$. Note that $n \in \emptyset$ is false, and thus $n \in \emptyset \implies n \subseteq \emptyset$ by vacuous truth. Since $\emptyset \in \mathbb{N}$, we have $\emptyset \in A$.

Now suppose that $m \in A$. Let $n \in s(m) = m \cup \{m\}$. By definition 3 and the [axiom schema of restricted comprehension](#), it follows that $n \in m$ or $n \in \{m\}$. If $n \in \{m\}$, then by an argument given previously, $n = m$. It suffices to show that $m \subseteq s(m)$. Indeed, if $k \in m$, then $k \in m$ or $k \in \{m\}$, hence $k \in m \cup \{m\} = s(m)$ by lemma 12. Thus, $m \subseteq s(m)$ by definition 4.

If $n \notin \{m\}$, then $n \in m$ by lemma 12. Since $m \in A$, by the [axiom schema of restricted comprehension](#), we have $n \in m$ implies $n \subseteq m$. Thus, $n \subseteq m$. By the above argument that $m \subseteq s(m)$, we have $n \subseteq m \subseteq s(m)$. By lemma 16, we have $n \subseteq s(m)$. Therefore, $s(m) \in A$, as all cases have been exhausted.

We now have $A \subseteq \mathbb{N}$ and $\emptyset \in A$ and $n \in A \implies s(n) \in A$. By Peano axiom (5), $A = \mathbb{N}$. Thus for any $m \in \mathbb{N}$, we have $m \in A$, and thus $n \in m$ implies $n \subseteq m$. ■

Lemma 19. *Let x be a set. Then $x \notin x$.*

Proof. Note that $\{x\}$ is nonempty, since $x \in \{x\}$. By the logic given in lemma 15, we also see that $a \in \{x\}$ if and only if $a = x$. By the axiom of regularity, since $\{x\}$ is nonempty, there is y such that $y \in \{x\}$, and there is no z with $z \in y$ and $z \in \{x\}$. Since $y \in \{x\}$ if and only if $y = x$, we have $y = x$. Thus there is no z with $z \in x$ and $z \in \{x\}$. If $x \in x$, then we have $x \in x$ and $x \in \{x\}$, a contradiction. Therefore, $x \notin x$. ■

(4) Let $n, m \in \mathbb{N}$ such that $s(n) = s(m)$. Note that $n \in s(n) = s(m)$ implies $n \in s(m) = m \cup \{m\}$. If $n \in \{m\}$, then $n = m$, and we are done. Suppose for contradiction that $n \in m$. Then we have $m \in s(m) = s(n)$ implies $m \in s(n) = n \cup \{n\}$. If $m \in \{n\}$, then $m = n$. Thus $n \in m = n$ implies $n \in n$, a contradiction by lemma 19. Thus we must take $m \in n$. By lemma 18, we have $m \subseteq n$. We now have $n \subseteq m$ and $m \subseteq n$, and therefore $m = n$. We thus have $n \in m = n$, and thus $n \in n$, a contradiction by lemma 19. ■

Define $0 := \emptyset$ to recover the presentation of the Peano axioms from class.